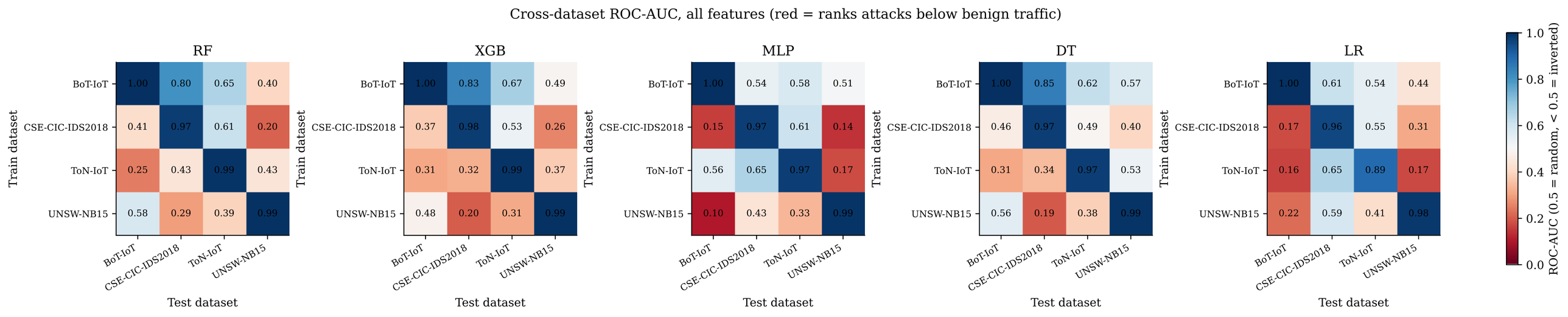
Cross-dataset ROC-AUC, all features (red = ranks attacks below benign traffic)
RF
XGB
MLP
DT
LR
Train dataset
Test dataset
BoT-IoT
CSE-CIC-IDS2018
ToN-IoT
UNSW-NB15
1.00 0.80 0.65 0.40
0.41 0.97 0.61 0.20
0.25 0.43 0.99 0.43
0.58 0.29 0.39 0.99
1.00 0.83 0.67 0.49
0.37 0.98 0.53 0.26
0.31 0.32 0.99 0.37
0.48 0.20 0.31 0.99
1.00 0.54 0.58 0.51
0.15 0.97 0.61 0.14
0.56 0.65 0.97 0.17
0.10 0.43 0.33 0.99
1.00 0.85 0.62 0.57
0.46 0.97 0.49 0.40
0.31 0.34 0.97 0.53
0.56 0.19 0.38 0.99
1.00 0.61 0.54 0.44
0.17 0.96 0.55 0.31
0.16 0.65 0.89 0.17
0.22 0.59 0.41 0.98
ROC-AUC (0.5 = random, < 0.5 = inverted)
1.0
0.8
0.6
0.4
0.2
0.0